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ELECTRONIC APPARATUS

Abstract

An electronic apparatus includes a display panel having a display surface and a rear surface on a side opposite to the display surface, a chassis member configured to support the rear surface of the display panel on an inner surface of the chassis member and having a standing wall having a receiving surface formed on an edge portion facing the display panel, a glass plate configured to cover the display surface of the display panel and having an outer edge portion that protrudes from an outer peripheral side surface of the display panel, in which an edge surface of the outer edge portion faces the standing wall with a gap, and a resin member forming a protruding portion on a back side of the outer edge portion of the glass plate without being interposed in the gap between the standing wall and the edge surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Japanese Patent Application No. JP 2024-029879 filed on Feb. 29, 2024, the contents of which are hereby incorporated herein by reference in their entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to an electronic apparatus including a display panel.

Description of the Related Art

[0003] An electronic apparatus such as a laptop PC includes a display panel configured with a liquid crystal display or the like. For example, Japanese Unexamined Patent Application Publication No. 2023-005541 discloses a configuration in which a display panel is supported by the inner surface of a chassis member having standing walls on the edge portions thereof.

SUMMARY OF THE INVENTION

[0004] In the configuration of Japanese Unexamined Patent Application Publication No. 2023-005541, a display surface of a display panel is covered with a glass plate. In this configuration, a bezel member made of a resin is interposed between the edge surface of the glass plate and the standing wall. The bezel member can suppress the glass plate from colliding with the standing wall and being broken or damaged when the display panel is attached to the chassis member or when the electronic apparatus is dropped.

[0005] Incidentally, it may be desired that the bezel width surrounding the outer circumference of the display panel be configured to be as narrow as possible to improve the appearance quality. In the configuration of Japanese Unexamined Patent Application Publication No. 2023-005541, the bezel width is increased by the thickness of the bezel member interposed between the glass plate and the standing wall. On the other hand, in a case in which the bezel member is eliminated from this configuration, there is a concern that the glass may directly interfere with the standing wall and be broken or damaged.

[0006] The present invention has been made in consideration of the problems in the related art described above, as well as other problems not described that will be appreciated by those of ordinary skill in the art, and aims to at least provide an electronic apparatus that can narrow the bezel width while suppressing the glass plate from being broken or damaged.

[0007] An electronic apparatus according to an aspect of the present invention includes a display panel having a display surface and a rear surface on a side opposite to the display surface, a chassis member configured to support the rear surface of the display panel on an inner surface of the chassis member and having a standing wall having a receiving surface formed on an edge portion facing the display panel, a glass plate configured to cover the display surface of the display panel and having an outer edge portion that protrudes from an outer peripheral side surface of the display panel, in which an edge surface of the outer edge portion faces the standing wall with a gap, and a resin member forming a protruding portion on a back side of the outer edge portion of the glass plate without being interposed in the gap between the standing wall and the edge surface, with the protruding portion locked by the receiving surface.

[0008] According to the above-described aspect of the present invention, it is possible to narrow the bezel width while suppressing the glass plate from being broken or damaged.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a schematic plan view of an electronic apparatus according to one or more embodiments as viewed from above.

[0010] FIG. **2** is a schematic front view of a chassis member.

[0011] FIG. **3** is a rear view of a display assembly.

[0012] FIG. **4** is a schematic side cross-sectional view of a first chassis taken along a line IV-IV in FIG. **2**.

[0013] FIG. **5** is a schematic side cross-sectional view of the first chassis taken along a line V-V in FIG. **2**.

[0014] FIG. **6** is a schematic perspective view showing an enlarged view of a bulging portion and a peripheral portion thereof.

[0015] FIG. **7** is a schematic cross-sectional view showing an operation of attaching a display assembly to a chassis member.

[0016] FIG. **8** is a schematic cross-sectional view of the first chassis including a stepped portion.

[0017] FIG. **9** is a schematic front view of a chassis member that does not have the bulging portion.

[0018] FIG. **10** is a rear view of the display assembly that can be attached to the chassis member shown in FIG. **9**.

DETAILED DESCRIPTION OF THE INVENTION

[0019] Embodiments of an electronic apparatus according to the present invention will be described in detail below with reference to the accompanying drawings.

[0020] FIG. **1** is a schematic plan view of an electronic apparatus **10** according to one or more embodiments as viewed from above. The electronic apparatus **10** has a configuration in which a first chassis **11** and a second chassis **12** are connected to each other by a hinge **14** to be rotationally movable relative to each other.

[0021] The electronic apparatus **10** is a so-called convertible type PC that can be used as a laptop PC (laptop mode) or a tablet type PC (tablet mode) depending on an angular position between the chassis **11** and **12**. The electronic apparatus **10** can set the chassis **11** and **12** at a desired angular position within an angular range of **0** degrees to **360** degrees. The **0** degrees is a position at which the surface directions of the chassis **11** and **12** are parallel to each other and a front surface **11***a* of the first chassis **11** and the surface (keyboard **16**) of the second chassis **12** face each other. The **360** degrees is a position at which the surface directions of the chassis **11** and **12** are parallel to each other and a rear surface **11***b* of the first chassis **11** and the bottom surface (surface opposite to the keyboard **16** side) of the second chassis **12** face each other. The electronic apparatus **10** may be various electronic computers such as a general-purpose laptop PC or a tablet type PC having a single plate shape, which can be set at a desired angular position within an angular range of approximately 0 degrees to 180 degrees between the chassis **11** and **12**, in addition to the convertible type PC.

[0022] The second chassis **12** is a flat box having a rectangular outer shape in a plan view, and is adjacent to the first chassis **11**. Inside the second chassis **12**, various electronic components such as a motherboard on which a CPU and the like are mounted, a battery device, a memory, an antenna device are accommodated. The keyboard **16** and a touch pad **17** face the upper surface of the second chassis **12**.

[0023] The first chassis **11** has a rectangular outer shape in a plan view and is a flat box that is thinner than the second chassis **12**. A display panel **18** is mounted on the first chassis **11**.

[0024] Hereinafter, the first chassis **11** and each of the components mounted thereon will be described based on the direction as seen by a user viewing a display surface **18***a* of the display panel **18**, with the left-right direction respectively referred to as X**1** and X**2** directions, the up-down direction respectively referred to as Y**1** and Y**2** directions, and the depth direction respectively referred to as Z**1** and Z**2** directions. The X**1** and X**2** directions are sometimes collectively referred

to as the X direction, and the Y**1** and Y**2** directions and the Z**1** and Z**2** directions are sometimes similarly referred to as the Y direction and the Z direction.
[0025] The display surface **18***a* of the display panel **18** faces the Z**1**-side surface (front surface **11***a*) of the first chassis **11**. The display surface **18***a* is the surface on which the video or image is displayed. The display panel **18** is configured with, for example, a liquid crystal display or an organic EL display. The display panel **18** has a structure in which, for example, glass, a liquid crystal layer, a light guide plate, or the like are laminated, and the outer peripheral edge portions of each layer are fixed to each other with a double-sided tape, an adhesive, or the like.
[0026] The display surface **18***a* of the display panel **18** is covered with a glass plate **19**. The glass plate **19** is a cover glass that covers substantially the entire front surface **11***a* of the first chassis **11** including the display panel **18**. The glass plate **19** can be configured with touch glass corresponding to a touch operation. The glass plate **19** may not support a touch operation.
[0027] The first chassis **11** has a chassis member **20**. FIG. **2** is a schematic front view of the chassis member **20**. FIG. **2** shows an inner surface **20***a* (back surface of the rear surface **11***b*) of the chassis member **20** and predetermined components installed on the inner surface **20***a*, and FIG. **9** described below is also the same.
[0028] The chassis member **20** has a rectangular plate portion **21** and a standing wall **22** standing upright from the outer peripheral edge portion of the plate portion **21**. The chassis member **20** has a shallow bathtub shape. The plate portion **21** forms the Z**2**-side surface (rear surface **11***b*) of the first chassis **11**. The standing wall **22** is provided to surround most of the four sides of the plate portion **21** and stands in the Z**1** direction. The chassis member **20** is made of, for example, metal and is formed of, for example, an aluminum alloy or a magnesium alloy. The chassis member **20** may be made of a resin. The chassis member **20** may be configured by joining a frame material that constitutes the standing wall **22** to the periphery of the plate portion **21** formed of, for example, a carbon fiber reinforced resin.
[0029] The chassis member **20** supports a rear surface **18***b* on a side opposite to the display surface **18***a* of the display panel **18** with the inner surface **20***a*. Specifically, the rear surface **18***b* of the display panel **18** is fixed to the inner surface **20***a* of the plate portion **21** by using a double-sided adhesive tape **24**.
[0030] The hinge **14** is connected to the Y**2**-side edge portion (edge portion **11***c*) of the first chassis **11** in a pair of left and right directions. A reference numeral **14***a* in FIG. **2** denotes a hinge fixing portion to which the hinge **14** is fixed. In a case in which the chassis member **20** is made of metal, the hinge fixing portion **14***a* can form a screw thread by direct processing on the plate portion **21**. In a case where the chassis member **20** is made of a resin, the hinge fixing portion **14***a* can be formed of a metal bracket to which the hinge **14** can be screwed. The chassis member **20** has a cutout portion **20***b* in which the hinge **14** is disposed on a side portion of the hinge fixing portion **14***a*.
[0031] The display panel **18** can be used as a display assembly **26** in which the glass plate **19** is fixed to the display surface **18***a*. As described above, the rear surface **18***b* of the display panel **18** is fixed to the chassis member **20** by the double-sided adhesive tape **24**. In this case, the outer periphery of the glass plate **19** extends along the inner periphery of the standing wall **22**. As a result, the display assembly **26** is attached to the chassis member **20**.
[0032] Next, an assembly structure of the display assembly **26** with respect to the chassis member **20** will be described.
[0033] FIG. **3** is a rear view of the display assembly **26**. FIG. **4** is a schematic side cross-sectional view of the first chassis **11** taken along the line IV-IV in FIG. **2**. FIG. **5** is a schematic side cross-sectional view of the first chassis **11** taken along the line V-V in FIG. **2**.
[0034] As shown in FIGS. **2**, **4**, and **5**, the standing wall **22** has a first portion **28** and a second portion **29**.
[0035] The first portion **28** is a portion provided at an upper portion (distal end side) of the standing

wall **22** in the standing direction (Z**1** direction) from the plate portion **21**. The first portion **28** has an interior wall surface **28***a* facing the inside of the first chassis **11**. The interior wall surface **28***a* faces an edge surface **19***b* of the glass plate **19** with a predetermined gap C therebetween.
[0036] The second portion **29** is a portion that is provided below the first portion **28** in the standing direction of the standing wall **22**. The plate thickness of the second portion **29** is larger than the plate thickness of the first portion. The second portion **29** is disposed to be under an outer edge portion **19***a* of the glass plate **19**, which will be described later. The second portion **29** has an interior wall surface **29***a* facing the inside of the first chassis **11** and a support surface **29***b* facing the Z**1** direction. The interior wall surface **29***a* faces an outer peripheral side surface **18***c* of the display panel **18** with a predetermined gap therebetween. The support surface **29***b* is a surface that protrudes from a lower end of the interior wall surface **28***a* of the first portion **28**. The support surface **29***b* can support a resin member **42** described later.
[0037] The standing wall **22** has a pair of horizontal walls **30** and **31** extending in the X direction and a pair of vertical walls **32** and **33** extending in the Y direction. As a result, the standing wall **22** extends to surround the entire periphery of the first chassis **11**. The second portion **29** is provided on at least the horizontal wall **30** on the Y**1** side, but may be similarly provided on the other walls **31** to **33**.
[0038] One horizontal wall **30** constitutes the Y**1**-side edge portion (edge portion **11***d*) of the first chassis **11**. The horizontal wall **30** extends over substantially the entire length of the edge portion **11***d*. The first chassis **11** has a bulging portion **34** that bulges outward (Y**1** side) at a central portion of the edge portion **11***d* in the longitudinal direction. The bulging portion **34** expands a space between the standing wall **22** (horizontal wall **30**) and the outer peripheral side surface **18***c* of the display panel **18**, and secures an installation space for a high-performance camera module **36**. The camera module **36** includes a camera lens **36***a*. The camera lens **36***a* is a camera that can be used for video capturing or the like, has a lens for focusing in the center, and has an image sensor provided on a back side of the lens. The camera module **36** can further include an IRLED (infrared camera), an illuminance sensor, and the like.
[0039] A pair of left and right recessed portions **38** and **38** and a pair of left and right engaged portions **39** and **39** are formed on the support surface **29***b* of the horizontal wall **30**.
[0040] The recessed portion **38** is a hole formed in the support surface **29***b*. The recessed portion **38** has, for example, a flat elliptical shape extending in the X direction (see FIG. **2**). A wall surface of the recessed portion **38** facing the display panel **18** side (Y**1** side) forms a receiving surface **38***a* that locks a protruding portion **48** of a resin member **42** described later. The engaged portion **39** has a shape in which the second portion **29** of the horizontal wall **30** is cut out in the Y**1** direction. The engaged portion **39** is a short plate piece that protrudes in the Y**2** direction from a lower end of the interior wall surface **28***a* of the first portion **28**.
[0041] The other horizontal wall **31** constitutes the edge portion **11***c* of the first chassis **11**. The horizontal wall **31** may have cutouts or the like at various locations. As a result, the horizontal wall **31** can form an installation space of a wiring line that connects the components of the hinge **14** and the chassis **11** and **12**.
[0042] The vertical walls **32** and **33** extend along the Y direction and are perpendicular to the horizontal walls **30** and **31**, respectively. As a result, corner wall portions CW**1** to CW**4** are formed at the four corners of the standing wall **22**.
[0043] Although details will be described later, it is preferable that the left and right recessed portions **38** are respectively disposed at positions as close to the corner wall portions CW**1** and CW**2** as possible. As shown in FIG. **2**, the horizontal wall **30** has a recess **24***a* on a side of the corner wall portions CW**1** and CW**2**. The recess **24***a* is formed by cutting out a part of the support surface **28***b* in the Y**1** direction. The recess **24***a* can be used to dispose the end portions of the double-sided adhesive tape **24**. In the horizontal wall **30**, a recess **37** in which the support surface **29***b* is cut out in an arc shape in the Y**1** direction is further formed between the recess **24***a* and the

corner wall portions CW**1** and CW**2**. The recess **37** is an escape portion for preventing the corners of the display panel **18** from colliding with the standing wall **22** due to an impact when the electronic apparatus **10** is dropped on a floor surface or the like. The left and right recessed portions **38** are disposed at positions as far as possible from the corner wall portions CW**1** and CW**2** at positions at which the recesses **24***a* and **37** do not interfere with the recessed portions **38**.
[0044] As shown in FIGS. **3** to **5**, the display assembly **26** is a laminate in which the glass plate **19** is fixed to the display surface **18***a* of the display panel **18**.
[0045] The glass plate **19** can be fixed by an adhesive member **40** that is transparent to the display surface **18***a*. The surface area of the glass plate **19** is slightly larger than the surface area of the display panel **18**. Therefore, the glass plate **19** has an eave-like outer edge portion **19***a* that protrudes outward from the outer peripheral side surface **18***c* of the display panel **18**.
[0046] In the glass plate **19**, the edge surface **19***b* of the outer edge portion **19***a* faces the interior wall surface **28***a* of the standing wall **22** with the gap C therebetween. The gap C is, for example, 0.2 mm. Specifically, in the four-sided standing wall **22**, the horizontal wall **30** on the Y**1** side and the left and right vertical walls **32** and **33** face the edge surface **19***b* with the gap C between the interior wall surface **28***a* of the first portion **28** and the edge surface **19***b*. Since the hinge **14** or the components related to the wiring line are disposed between the edge surface **19***b* and the horizontal wall **31** on the Y**2** side, the gap between the interior wall surface **28***a* and the edge surface **19***b* is wide, for example, 10 mm or more.
[0047] Corner portions C**1** to C**4** that are curved along the corner wall portions CW**1** to CW**4** of the chassis member **20** are formed at four corners of the glass plate **19**. The corner portions C**1** and C**2** provided at one end and the other end of the outer edge portion **19***a* on the Y**1** side in the longitudinal direction face the interior wall surfaces **28***a* of the corner wall portions CW**1** and CW**2** with the gap C therebetween. The corner portions C**3** and C**4** provided at both ends of the outer edge portion **19***a* on the Y**2** side are disposed with the same gap as the wide gap (for example, 10 mm or more) between the corner wall portions CW**3** and CW**4** and the horizontal wall **31** described above.
[0048] The display assembly **26** further includes a pair of left and right resin members **42** and **42** and a bezel member **44**. The resin member **42** and the bezel member **44** can be fixed to a back surface **19***c* of the outer edge portion **19***a* of the glass plate **19** on the Y**1** side by using an adhesive member **46**. The adhesive member **46** can be exemplified by a double-sided adhesive tape, an adhesive, or a bonding agent. Hereinafter, the outer edge portion **19***a* of the glass plate **19** that extends in the X direction on the Y**1** side may be referred to as an “outer edge portion **19***a***1**”. A bulging portion **19***a***2** corresponding to the shape of the bulging portion **34** is formed on the outer edge portion **19***a***1** on the Y**1** side at a central portion in the longitudinal direction.
[0049] As shown in FIGS. **3** to **5**, the resin member **42** is a strip-shaped resin part extending in the X direction. The resin member **42** on the X**1** side extends from a position close to the corner portion C**1** to the front of the bulging portion **19**a**2**. The resin member **42** on the X**2** side extends from a position close to the corner portion C**2** to the front of the bulging portion **19**a**2**. Each resin member **42** is fixed to the back surface **19***c* and is not interposed in the gap C between the interior wall surface **28***a* of the standing wall **22** and the edge surface **19***b* of the glass plate **19**. The resin member **42** is disposed on the support surface **29***b* of the horizontal wall **30** with a slight gap.
[0050] Each resin member **42** has the protruding portion **48** and a hook **50**.
[0051] The protruding portion **48** is a rib that protrudes downward (Z**2** direction) on the back surface **19***c* side of the glass plate **19**. The protruding portion **48** has, for example, a flat elliptical shape extending in the X direction (see FIG. **3**). The protruding portion **48** has a size that allows the protruding portion **48** to be inserted into the recessed portion **38** on the standing wall **22** side. In a state in which the display assembly **26** is attached to the chassis member **20**, the protruding portions **48** of the respective resin members **42** are inserted into the left and right recessed portions **38**. In this state, the protruding portion **48** abuts against the receiving surface **38***a* of the recessed

portion **38** at a stopper surface **48***a*. The stopper surface **48***a* is a surface facing a side (Y**1** side) opposite to the display panel **18** side. As a result, the protruding portion **48** restricts the movement of the display assembly **26** in the Y**1** direction. That is, the protruding portion **48** can prevent the edge surface **19***b* of the outer edge portion **19***a***1** from coming into contact with the interior wall surface **28***a*.

[0052] The hook **50** is a claw-like protrusion. The hook **50** protrudes downward on the back surface **19***c* side of the glass plate **19** and then is bent toward the standing wall **22** side (Y**1** side). The hook **50** can be disengaged from the engaged portion **39** on the standing wall **22** side. As a result, the hook **50** prevents the display assembly **26** from falling off from the chassis member **20** in the Z**1** direction.

[0053] FIG. **6** is an enlarged schematic perspective view of the bulging portion **34** and a peripheral portion thereof.

[0054] As shown in FIGS. **3** and **6**, the bezel member **44** is a strip-shaped resin part extending in the X direction. The bezel member **44** is provided on the back surface **19***c* of the glass plate **19** between the left and right resin members **42** and **42**. The bezel member **44** has a plate portion **44***a* and a bezel portion **44***b*, and has a substantially L-shaped cross-sectional shape.

[0055] The plate portion **44***a* is fixed to the back surface **19***c* of the bulging portion **19***a***2** that covers the bulging portion **34** of the chassis member **20**. The plate portion **44***a* is provided to cover the upper side (Z**1** side) of the camera module **36**.

[0056] The bezel portion **44***b* is a thin plate having a standing wall shape that stands up in the Z**1** direction from an edge portion of the plate portion **44***a* on the Y**1** side. The bezel portion **44***b* is interposed in a gap Cb between the interior wall surface **28***a* of the standing wall **22** located at the bulging portion **34** and the edge surface **19***b* of the glass plate **19** (see FIG. **6**). The gap Cb is larger than the gap C described above and is required to be, for example, about 0.5 mm to 1 mm.

[0057] The camera shutter **52** is supported by the bezel member **44** to be slidable in the X direction. The camera shutter **52** is a shutter member that is capable of opening and closing at least the camera lens **36***a*. The camera shutter **52** includes a shutter portion **52***a* and an operation knob **52***b*.

[0058] The shutter portion **52***a* is slidable in the X direction on the lower side (Z**2** side) of the plate portion **44***a*. The operation knob **52***b* stands up from the Y**1**-side edge portion of the shutter portion **52***a* in the Z**1** direction. The operation knob **52***b* can be slid in the X direction by hand. The camera shutter **52** can be switched between the closed position and the open position by moving the operation knob **52***b* in the X direction. The closed position is a position at which the shutter portion **52***a* covers the camera lens **36***a*. The open position is a position at which the shutter portion **52***a* does not cover the camera lens **36***a*.

[0059] As shown in FIG. **6**, the bezel portion **44***b* has a defective portion **44***b***1** in which a part of the bezel portion **44***b* in the longitudinal direction is cut out. The operation knob **52***b* is disposed in the defective portion **44***b***1** and is movable in the X direction in the defective portion **44***b***1**. As a result, the operation knob **52***b* appears to be a part of the bezel portion **44***b* and is not noticeable in appearance.

[0060] As described above, in the electronic apparatus **10**, the bezel member is not interposed between the horizontal wall **30** excluding the bulging portion **34** and the glass plate **19**, between the horizontal wall **31** and the glass plate **19**, and between the vertical walls **32** and **33** and the glass plate **19**. Therefore, the bezel width between the horizontal wall **30** excluding at least the bulging portion **34** and the glass plate **19** and between the vertical walls **32** and **33** and the glass plate **19** is extremely narrow.

[0061] On the other hand, the bezel portion **44***b* of the bezel member **44** is interposed between the standing wall **22** positioned at the bulging portion **34** and the glass plate **19**. The bezel portion **44***b* can have the function of preventing the operation knob **52***b* from being noticeable in appearance. Further, the bezel portion **44***b* can also have a function of preventing the operation knob **52***b* protruding from the surface of the glass plate **19** from being broken.

[0062] The electronic apparatus **10** can be configured not to have the camera shutter **52**. The electronic apparatus **10** can also have a configuration in which the operation knob **52***b* does not protrude from the surface (front surface **11***a*) of the glass plate **19**. In a case of these configurations, the bezel member **44** can be configured with a resin part that does not have the bezel portion **44***b*, or the bezel member **44** itself can be omitted.

[0063] Next, an operation of attaching the display assembly **26** to the chassis member **20** and an action and effect of the electronic apparatus **10** will be described. FIG. **7** is a schematic cross-sectional view showing an operation of attaching the display assembly **26** to the chassis member **20**.

[0064] As shown in FIG. **7**, the display assembly **26** is in a posture in which the outer edge portion **19***a***1** of the glass plate **19** is slightly lowered, and the stopper surface **48***a* of the protruding portion **48** abuts against the receiving surface **38***a* of the recessed portion **38**. In this case, in the display assembly **26**, the stopper surface **48***a* abuts against the receiving surface **38***a* before the edge surface **19***b* of the outer edge portion **19***a***1** comes into contact with the interior wall surface **28***a* of the horizontal wall **30**. When the stopper surface **48***a* abuts against the receiving surface **38***a*, the hook **50** is engaged with the engaged portion **39** substantially at the same time. Before this process, the release paper around the edge portion on the Y**1** side of the double-sided adhesive tape **24** is peeled off to expose the adhesive surface with respect to the rear surface **18***b* of the display panel **18**.

[0065] In this case, the display assembly **26** is positioned in the Y direction with respect to the chassis member **20** at the Y**1**-side edge portion. Further, a part of the display panel **18** is fixed to the inner surface **20***a* of the chassis member **20** by the double-sided adhesive tape **24**. Therefore, the release paper of the double-sided adhesive tape **24** is completely peeled off, and the entire display assembly **26** is lowered.

[0066] As a result, the display assembly **26** is attached to the chassis member **20** in a state in which the entire display assembly **26** is positioned in the XY direction.

[0067] As described above, the electronic apparatus **10** can include the chassis member **20** that supports the rear surface **18***b* of the display panel **18** with the inner surface **20***a* and has the standing wall **22** in which the receiving surface **38***a* facing the direction of the display panel **18** is formed at the edge portion. The electronic apparatus **10** can further include the glass plate **19** having the outer edge portion **19***a* that protrudes from the outer peripheral side surface **18***c* of the display panel **18** and the edge surface **19***b* that faces the standing wall **22** with the gap C therebetween. The electronic apparatus **10** can further include the resin member **42** that forms the protruding portion **48** on the back surface **19***c* side of the outer edge portion **19***a* of the glass plate **19** without being interposed in the gap C between the standing wall **22** and the edge surface **19***b*, with the protruding portion **48** locked by the receiving surface **38***a*.

[0068] As described above, in the electronic apparatus **10**, in a state in which the stopper surface **48***a* abuts against the receiving surface **38***a*, the gap C is secured between the glass plate **19** and the standing wall **22** (see FIG. **4**). Therefore, in the electronic apparatus **10**, when the display assembly **26** is attached to the chassis member **20**, the stopper surface **48***a* abuts against the receiving surface **38***a* before the edge surface **19***b* comes into contact with the interior wall surface **28***a*. As a result, in the electronic apparatus **10**, it is possible to suppress the edge surface **19***b* of the glass plate **19** from colliding with the standing wall **22** and being broken or damaged by mistake when the display assembly **26** is attached.

[0069] Further, when the electronic apparatus **10** is accidentally dropped on a floor surface or the like, it is possible to suppress the glass plate **19** from being broken or the like even in a case in which the impact is applied to the corner wall portions CW**1**, CW**2**, and the like. This is because the stopper surface **48***a* and the receiving surface **38***a* abut with each other, and thus the interior wall surface **28***a* of the standing wall **22** can be prevented from colliding with the edge surface **19***b* of the glass plate **19** in the electronic apparatus **10**.

[0070] Moreover, when at least the bulging portion **34** is removed, the bezel member is not interposed between the edge surface **19***b* of the glass plate **19** and the interior wall surface **28***a* of the standing wall **22** in the electronic apparatus **10**. As a result, the gap C can be extremely narrowed to, for example, 0.2 mm. The gap C is significantly smaller than the gap Cb in which the bezel portion **44***b* is interposed as described above. As a result, the electronic apparatus **10** can narrow the width (bezel width) between the display assembly **26** and the standing wall **22**, and the appearance quality is improved.

[0071] As shown in FIGS. **2** and **3**, the protruding portion **48** and the receiving surface **38***a* can be respectively provided at positions close to the corner portions C**1** and C**2** at both ends of the outer edge portion **19***a***1**, respectively. In this case, even in a case in which the display assembly **26** is in a posture inclined to some extent in the surface direction (XY direction) when the display assembly **26** shown in FIG. **7** is attached, one of the left and right protruding portions **48** abuts against the receiving surface **38***a* reliably. Therefore, it is possible to further suppress the glass plate **19** from being broken or the like when the display assembly **26** is attached.

[0072] The receiving surface **38***a* may have a configuration other than the configuration of the recessed portion **38** as long as the receiving surface **38***a* can receive the protruding portion **48** of the display assembly **26**. FIG. **8** is a schematic cross-sectional view of the first chassis **11** including a stepped portion **54** instead of the recessed portion **38**.

[0073] The stepped portion **54** shown in FIG. **8** forms the receiving surface **38***a* instead of the recessed portion **38**. The stepped portion **54** is not a hole formed in the support surface **29***b* as in the recessed portion **38**, but is a stepped portion obtained by cutting out a part of the second portion **29** of the horizontal wall **30** from the interior wall surface **28***a* side toward the Y**1** side. A surface of the stepped portion **54** facing the display panel **18** side (Y**1** side) also forms the receiving surface **38***a*.

[0074] Therefore, even in the configuration in which the stepped portion **54** is used, the stopper surface **48***a* of the protruding portion **48** on the display assembly **26** side is locked by the receiving surface **38***a*, and it is possible to suppress the glass plate **19** from being broken or the like.

[0075] As shown in FIG. **8**, in the configuration using the stepped portion **54**, a protruding portion **48**A can also be used instead of the rib-shaped protruding portion **48** formed on the resin member **42**. The protruding portion **48**A has a configuration in which substantially the entire resin member **42** protrudes downward, and a Y**1** side surface thereof is the stopper surface **48***a*. In other words, the protruding portion **48**A is formed by using the thickness of the resin member **42** as it is.

[0076] In the above description, the configuration in which the first chassis **11** includes the bulging portion **34** has been exemplified, but the bulging portion **34** can also be omitted. FIG. **9** is a schematic front view of a chassis member **20**A that does not have the bulging portion **34**. FIG. **10** is a rear view of the display assembly **26**A that can be attached to the chassis member **20**A shown in FIG. **9**.

[0077] The chassis member **20**A shown in FIG. **9** does not have the bulging portion **34**. Therefore, the chassis member **20**A has the standing wall **22** (horizontal wall **30**) along the edge portion **11***d* that extends linearly over the entire length. The recessed portion **38** and the engaged portion **39** are also formed in the horizontal wall **30** of the chassis member **20**A. The camera module **36** can be disposed in a space obtained by resecting, for example, the second portion **29** in a range including the center of the horizontal wall **30** in the longitudinal direction.

[0078] The display assembly **26**A shown in FIG. **10** does not have the bulging portion **19**a**2**. Therefore, in the glass plate **19** of the display assembly **26**A, the outer edge portion **19***a***1** extends linearly over the entire length. In the display assembly **26**A, the bezel member **44** can be omitted.

[0079] In the first chassis **11** using the chassis member **20**A and the display assembly **26**A, the narrow gap C can be formed between the glass plate **19** and the standing wall **22** over the entire length of the edge portion **11***d*. The bezel member **44** can also be used in the display assembly **26**A.

[0080] In the display assembly **26**A shown in FIG. **10**, for example, one long resin member **42**A

that extends along the substantially entire length of the outer edge portion **19***a***1** can also be used instead of the resin member **42**. Since the pair of protruding portions **48** (**48**A) and the pair of hooks **50** can be formed in the resin member **42**A, the number of components can be reduced. In addition, a pair of left and right resin members **42** can also be used for the display assembly **26**A.
[0081] The present invention is not limited to the above-described embodiments, and it goes without saying that the present invention can be freely modified without departing from the gist of the present invention.
DESCRIPTION OF SYMBOLS
[0082] **10** electronic apparatus [0083] **11** first chassis [0084] **12** second chassis [0085] **14** hinge [0086] **18** display panel [0087] **19** glass plate [0088] **19***a*, **19***a***1** outer edge portion [0089] **19***b* edge surface [0090] **20**, **20**A chassis member [0091] **22** standing wall [0092] **26**, **26**A display assembly [0093] **28***a*, **29***a* interior wall surface [0094] **36***a* camera lens [0095] **38** recessed portion [0096] **38***a* receiving surface [0097] **42**, **42**A resin member [0098] **44** bezel member [0099] **48**, **48**A protruding portion [0100] **48***a* stopper surface [0101] **52** camera shutter [0102] **54** stepped portion [0103] C**1** to C**4** corner portion [0104] CW**1** to CW**4** corner wall portion

Claims

1. An electronic apparatus comprising: a display panel having a display surface and a rear surface on a side opposite to the display surface; a chassis member configured to support the rear surface of the display panel on an inner surface of the chassis member and having a standing wall having a receiving surface formed on an edge portion facing the display panel; a glass plate configured to cover the display surface of the display panel and having an outer edge portion that protrudes from an outer peripheral side surface of the display panel, in which an edge surface of the outer edge portion faces the standing wall with a gap; and a resin member forming a protruding portion on a back side of the outer edge portion of the glass plate without being interposed in the gap between the standing wall and the edge surface, with the protruding portion locked by the receiving surface.
2. The electronic apparatus according to claim 1, wherein the standing wall has a first portion that is provided at an upper portion in a standing direction of the standing wall and faces an edge surface of the glass plate, and a second portion that is provided below the first portion to protrude toward a display panel side and is disposed below the resin member, the second portion has a recessed portion or a stepped portion, and one wall surface of the recessed portion or the stepped portion forms the receiving surface.
3. The electronic apparatus according to claim 2, wherein the glass plate has a pair of corner portions provided at one end and the other end of the outer edge portion in a longitudinal direction, and the protruding portion and the receiving surface are provided at positions closer to the pair of corner portions than a center in a direction along the longitudinal direction of the outer edge portion, respectively.
4. The electronic apparatus according to claim 3, wherein a pair of the resin members are provided, and the protruding portion is provided on each of the pair of resin members.
5. The electronic apparatus according to claim 4, wherein the chassis member has a bulging portion in which the standing wall in a range facing a portion including a center of an outer edge portion of the glass plate in a longitudinal direction bulges in a direction away from the display panel, the electronic apparatus further includes a camera lens disposed in the bulging portion, a camera shutter that includes an operation knob and is capable of opening and closing the camera lens, and a bezel member that is interposed between the standing wall positioned in the bulging portion and the edge surface of the glass plate, and the bezel member has a defective portion that extends along a longitudinal direction of the standing wall positioned in the bulging portion, with a part of the operation knob disposed to be movable.
6. The electronic apparatus according to claim 1, further comprising: a first chassis having the

chassis member and having a rectangular outer shape; a second chassis having a rectangular outer shape and adjacent to the first chassis; and a hinge configured to connect one edge portion of the first chassis and one edge portion of the second chassis, to be rotationally movable relative to each other, wherein the standing wall has a pair of vertical walls that extend along a direction perpendicular to the one edge portion of the first chassis to which the hinge is connected, and a horizontal wall that extends to be perpendicular to the pair of vertical walls on a side opposite to the one edge portion of the first chassis to which the hinge is connected, and the receiving surface is formed on the horizontal wall.